

Exercise Sheet #6**AC 1****Exercise 1: Sinusoidal Voltage**

Given the sinusoidal voltage $v(t) = 50 \cos(30t + 10^\circ)$ V, find:

- a) the amplitude V_m
- b) the period T
- c) the frequency f
- d) $v(t)$ at $t = 10$ ms.

Solution

- a) Amplitude:

$$V_m = 50 \text{ V}$$

- b) Period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{30} \approx 209.4 \text{ ms}$$

- c) Frequency:

$$f = \frac{1}{T} = \frac{30}{2\pi} \approx 4.775 \text{ Hz}$$

- d) Voltage at $t = 10$ ms:

$$v(0.01) = 50 \cos(30 \cdot 0.01 + 10^\circ)$$

Convert 10° to radians:

$$10^\circ = \frac{10\pi}{180} = 0.17453 \text{ rad}$$

Compute the angle:

$$30 \cdot 0.01 + 0.17453 = 0.47453 \text{ rad}$$

Thus:

$$v(0.01) = 50 \cos(0.47453) \approx 44.48 \text{ V}$$

Exercise 2: Phase Shift

Given $v_1 = 45 \sin(\omega t + 30^\circ)$ V and $v_2 = 50 \cos(\omega t - 30^\circ)$ V, determine the phase angle between the two sinusoids and which one lags the other.

Solution

Convert v_2 to sine form:

$$v_2 = 50 \cos(\omega t - 30^\circ) = 50 \sin(\omega t - 30^\circ + 90^\circ) = 50 \sin(\omega t + 60^\circ)$$

Thus the phase angles are

$$\phi_1 = 30^\circ, \quad \phi_2 = 60^\circ$$

Phase difference:

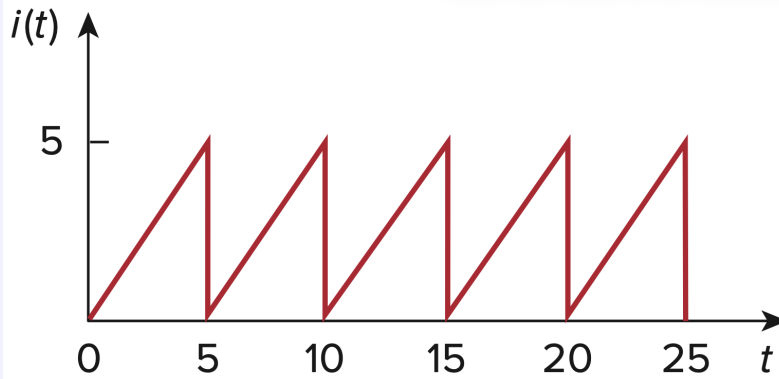
$$\Delta\phi = \phi_2 - \phi_1 = 60^\circ - 30^\circ = 30^\circ$$

Since $\phi_1 < \phi_2$,

v_1 lags v_2 by 30° .

Exercise 3: RMS Value

Calculate the RMS value of the current waveform below.



Solution

The waveform is a sawtooth with peak value 5 A and period $T = 5$ s:

$$I = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} = \sqrt{\frac{1}{5} \int_0^5 \left(\frac{5t}{5}\right)^2 dt} = \sqrt{\frac{1}{5} \cdot \frac{1}{3} \cdot (5^3 - 0^3)} \text{ A} \approx 2.89 \text{ A}$$

Exercise 4: Peak Phasors

For the following signals, find the complex amplitudes corresponding to a cosine reference as well as the rectangular form of the phasors.

- a) $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$
- b) $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

Solution

- a) Peak phasor (cosine reference):

$$\underline{\hat{v}} = 21 \angle -15^\circ \text{ V}$$

Rectangular form:

$$\underline{\hat{v}} = 21 (\cos(-15^\circ) + j \sin(-15^\circ)) \text{ V} \approx (20.28 - j5.44) \text{ V}$$

- b) Convert to cosine form:

$$i(t) = -8 \sin(10t + 70^\circ) \text{ mA} = 8 \sin(10t + 250^\circ) \text{ mA} = 8 \cos(10t + 160^\circ) \text{ mA}$$

Peak phasor (cosine reference):

$$\hat{\underline{i}} = 8 \angle 160^\circ \text{ mA}$$

Rectangular form:

$$\hat{\underline{i}} = 8 (\cos(160^\circ) + j \sin(160^\circ)) \text{ mA} \approx (-7.52 + j2.74) \text{ mA}$$

Exercise 5: Phasor Arithmetic

Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$.
If $v_1 = 10 \cos(50t - \frac{\pi}{3}) \text{ V}$ and $v_2 = 12 \cos(50t + 30^\circ) \text{ V}$, find v .

Solution

Convert both to phasors:

$$\hat{v}_1 = 10 \angle -60^\circ \text{ V}, \quad \hat{v}_2 = 12 \angle 30^\circ \text{ V}$$

Convert to rectangular form:

$$\hat{v}_1 = 10 (\cos(-60^\circ) + j \sin(-60^\circ)) \text{ V} = (5 - j8.66) \text{ V}$$

$$\hat{v}_2 = 12 (\cos(30^\circ) + j \sin(30^\circ)) \text{ V} = (10.39 + j6) \text{ V}$$

Sum the phasors:

$$\hat{v} = \hat{v}_1 + \hat{v}_2 = ((5 + 10.39) + j(-8.66 + 6)) \text{ V} = (15.39 - j2.66) \text{ V}$$

Convert back to polar form:

$$|\hat{v}| = \sqrt{15.39^2 + (-2.66)^2} \text{ V} \approx 15.62 \text{ V}$$

$$\theta = \tan^{-1} \left(\frac{-2.66}{15.39} \right) \approx -9.8^\circ$$

Thus:

$$\hat{v} = 15.62 \angle -9.8^\circ \text{ V}$$

Convert back to time domain:

$$v(t) = 15.62 \cos(50t - 9.8^\circ) \text{ V}$$

Exercise 6: AC Circuit

A series RLC circuit has $R = 80 \Omega$, $L = 240 \text{ mH}$, and $C = 5 \text{ mF}$.
If the input voltage is $v(t) = 10 \cos(2t) \text{ V}$, find the current flowing through the circuit.

Solution

Angular frequency:

$$\omega = 2 \text{ rad/s}$$

Impedances:

$$\underline{Z}_R = 80 \Omega$$

$$\underline{Z}_L = j\omega L = (j2 \cdot 0.24) \Omega = j0.48 \Omega$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{1}{j2 \cdot 0.005} \Omega = -j100 \Omega$$

Total impedance:

$$\underline{Z} = \underline{Z}_R + \underline{Z}_L + \underline{Z}_C = 80 \Omega + j0.48 \Omega - j100 \Omega = (80 - j99.52) \Omega$$

Magnitude and phase of total impedance:

$$|\underline{Z}| = \sqrt{80^2 + (-99.52)^2} \Omega \approx 127.69 \Omega$$

$$\theta_Z = \tan^{-1} \left(\frac{-99.52}{80} \right) \approx -51.21^\circ$$

Phasor form of voltage:

$$\underline{\hat{v}} = 10 \angle 0^\circ \text{ V}$$

Phasor form of current:

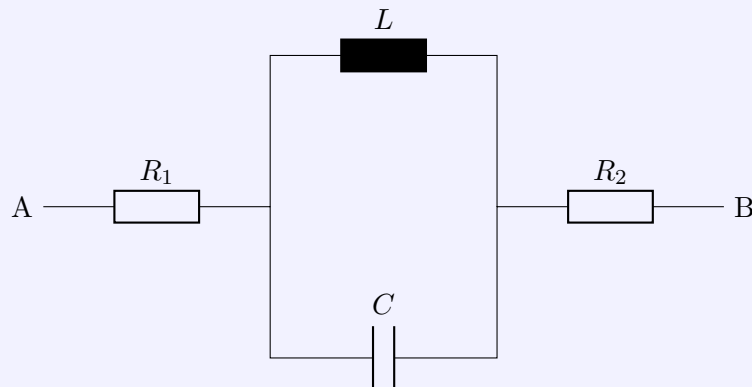
$$\underline{\hat{i}} = \frac{\underline{\hat{v}}}{\underline{Z}} = \frac{10 \angle 0^\circ \text{ V}}{127.69 \angle -51.21^\circ \Omega} = \frac{10 \text{ V}}{127.69 \exp(-j51.21^\circ) \Omega} \approx 78.3 \angle 51.21^\circ \text{ mA}$$

Convert back to time domain:

$$i(t) = 78.3 \cos(2t + 51.21^\circ) \text{ mA}$$

Exercise 7: Equivalent Impedance

Given the AC network with $R_1 = 10 \Omega$, $R_2 = 20 \Omega$, $L = 50 \text{ mH}$, $C = 100 \mu\text{F}$, $\omega = 1000 \text{ rad/s}$:



Determine:

1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Solution

$$\underline{Z}_R = R$$

$$\underline{Z}_L = j\omega L = j \cdot 1000 \cdot 0.05 = j50 \Omega$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 100 \cdot 10^{-6}} = -j10 \Omega$$

$$\underline{Z}_{LC} = \left(\frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C} \right)^{-1} = \frac{1}{j0.08} = -j12.5 \Omega$$

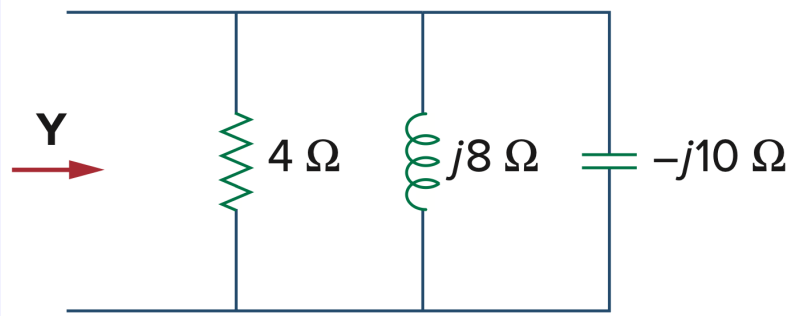
$$\underline{Z} = R_1 + \underline{Z}_{LC} + R_2 = (30 - j12.5) \Omega$$

$$|\underline{Z}| = \sqrt{30^2 + 12.5^2} \approx 32.5 \Omega$$

$$\varphi = \arctan \left(\frac{-12.5}{30} \right) \approx -22.6^\circ$$

Exercise 8: Equivalent Admittance

Determine the admittance $\underline{Y}$ for the circuit shown below.

**Solution**

Calculate the individual admittances:

$$\underline{Y}_R = \frac{1}{\underline{Z}_R} = \frac{1}{4 \Omega} = 0.25 \text{ S}$$

$$\underline{Y}_L = \frac{1}{\underline{Z}_L} = \frac{1}{j8 \Omega} = -j0.125 \text{ S}$$

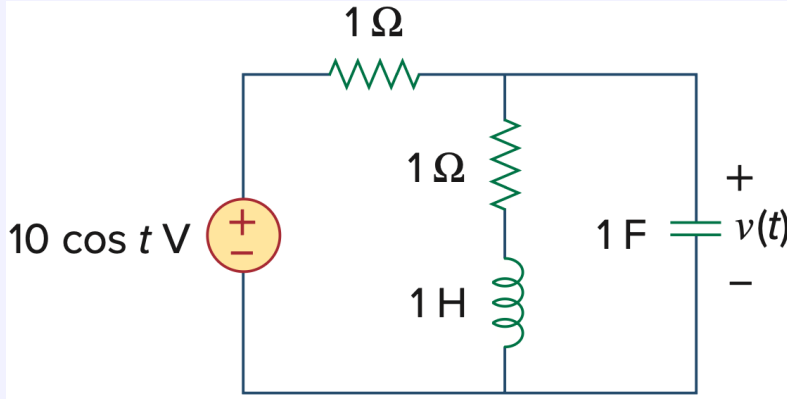
$$\underline{Y}_C = \frac{1}{\underline{Z}_C} = \frac{1}{-j10 \Omega} = j0.1 \text{ S}$$

Total admittance:

$$\underline{Y} = \underline{Y}_R + \underline{Y}_L + \underline{Y}_C = 0.25 \text{ S} - j0.125 \text{ S} + j0.1 \text{ S} = (250 - j25) \text{ mS}$$

Exercise 9: RLC Circuit

Find $v(t)$ in the circuit below.



Solution

We are given the time-domain source

$$v_s(t) = 10 \cos t \text{ V}.$$

In complex (exponential) representation we write this as the real part of a complex exponential:

$$v_s(t) = \text{Re}(\hat{v}_s e^{j\omega t}), \quad \hat{v}_s = 10, \quad \omega = 1 \text{ rad/s},$$

so $v_s(t) = \text{Re}(10e^{jt})$.

At $\omega = 1$,

$$\underline{Z}_{R_1} = 1, \quad \underline{Z}_{R_2} = 1, \quad \underline{Z}_L = j\omega L = j, \quad \underline{Z}_C = \frac{1}{j\omega C} = -j.$$

The series $R_2 - L$ branch impedance is

$$\underline{Z}_{RL} = 1 + j.$$

The parallel combination of $\underline{Z}_{RL}$ and $\underline{Z}_C$ has admittance

$$\underline{Y}_p = \frac{1}{\underline{Z}_{RL}} + \frac{1}{\underline{Z}_C} = \frac{1}{1+j} + \frac{1}{-j}.$$

Compute each term explicitly:

$$\frac{1}{1+j} = \frac{1-j}{(1+j)(1-j)} = \frac{1-j}{2} = \frac{1}{2} - \frac{1}{2}j, \quad \frac{1}{-j} = j.$$

Thus

$$\underline{Y}_p = \left(\frac{1}{2} - \frac{1}{2}j\right) + j = \frac{1}{2} + \frac{1}{2}j.$$

Therefore the equivalent impedance of the parallel network is

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \frac{1}{\frac{1}{2} + \frac{1}{2}j} = \frac{\frac{1}{2} - \frac{1}{2}j}{(\frac{1}{2})^2 + (\frac{1}{2})^2} = 1 - j.$$

The total series impedance seen by the source is

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_1} + \underline{Z}_p = 1 + (1 - j) = 2 - j.$$

Now form the complex current $\hat{i}$ (the complex amplitude multiplying e^{jt}) by dividing the complex source amplitude by the complex impedance:

$$\hat{i} = \frac{\hat{v}_s}{\underline{Z}_{\text{tot}}} = \frac{10}{2-j}.$$

Perform the algebraic division by multiplying numerator and denominator by the conjugate:

$$\hat{i} = 10 \frac{2+j}{(2-j)(2+j)} = 10 \frac{2+j}{5} = 4+2j.$$

Thus the complex current amplitude is $\hat{i} = 4+2j$. (Its magnitude is $|\hat{i}| = \sqrt{4^2 + 2^2} \approx 4.472$ and its angle $\arg(\hat{i}) = \arctan(2/4) = 26.57^\circ$, if one prefers polar form.)

The complex voltage across the parallel network (and thus across the capacitor) is

$$\hat{v}_p = \hat{i} \underline{Z}_p = (4+2j)(1-j).$$

Multiply out:

$$\hat{v}_p = 4(1-j) + 2j(1-j) = 4 - 4j + 2j - 2j^2 = 4 - 2j + 2 = 6 - 2j.$$

So the complex amplitude of the capacitor voltage is $\hat{v}_p = 6 - 2j$.

We may express this two ways:

(a) Time-domain real part of the exponential:

$$v(t) = \text{Re}(\hat{v}_p e^{jt}) = \text{Re}((6-2j)e^{jt}).$$

Expanding $(6-2j)e^{jt} = (6-2j)(\cos t + j \sin t)$ and taking the real part gives

$$v(t) = 6 \cos t + 2 \sin t.$$

(b) Amplitude-phase form: compute magnitude and angle of $\hat{v}_p$:

$$|\hat{v}_p| = \sqrt{6^2 + (-2)^2} \approx 6.325,$$

$$\arg(\hat{v}_p) = \arctan\left(\frac{-2}{6}\right) \approx -18.43^\circ.$$

Thus

$$\hat{v}_p = 6.325 \angle (-18.43^\circ),$$

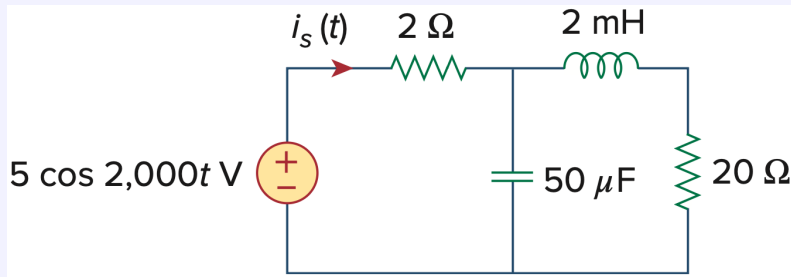
and therefore

$$v(t) = 6.325 \cos(t - 18.43^\circ) \text{ V},$$

which is algebraically equivalent to $v(t) = 6 \cos t + 2 \sin t$.

Exercise 10: Phase-shifted Source Current

Determine the value of $i_s(t)$ in the circuit below.



Solution

$$\omega = 2000 \text{ rad/s}$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{-j}{2000 \cdot 50 \times 10^{-6}} \Omega = -j10 \Omega$$

$$\underline{Z}_{RL} = 20 + j\omega L = (20 + j2000 \cdot 2 \times 10^{-3}) \Omega = (20 + j4) \Omega$$

$$\underline{Y}_p = \frac{1}{\underline{Z}_C} + \frac{1}{\underline{Z}_{RL}} = j0.1 + \frac{1}{20 + j4} = j0.1 + \frac{20 - j4}{416} = \left(\frac{20}{416} + j\frac{37.6}{416} \right) \text{ S}$$

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \left(\frac{\frac{20}{416} - j\frac{37.6}{416}}{\left(\frac{20}{416}\right)^2 + \left(\frac{37.6}{416}\right)^2} \right) \Omega = (4.58716 - j8.62385) \Omega$$

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_2} + \underline{Z}_p = 2 \Omega + (4.58716 - j8.62385) \Omega = (6.58716 - j8.62385) \Omega$$

$$\hat{i}_s = \frac{\hat{v}_s}{\underline{Z}_{\text{tot}}} = \frac{5}{6.58716 - j8.62385} \text{ A} = (0.27714 + j0.36949) \text{ A}$$

$$\hat{i}_s = \sqrt{(0.27714)^2 + (0.36949)^2} \text{ A} = 0.460753 \text{ A}$$

$$\varphi_{i_s} = \tan^{-1} \left(\frac{0.36949}{0.27714} \right) = 52.6263^\circ$$

$$i_s(t) = 460.7 \cos(2000t + 52.63^\circ) \text{ mA}$$